

## ASHEVILLE 13 S, NC, USA

WMO: 746970

Lat: 35.4185N

Lon: 82.5567W

Elev: 2103

StdP: 13.61

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53878

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 14.7       | 19.0 | 2.5                           | 6.8 | 20.8 | 6.1 | 8.1 | 24.6 | 12.8                  | 53.3 | 11.4 | 51.8 | 3.9                   | N/A  | 0.366 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.4                   | 86.7            | 70.4 | 84.7 | 70.1 | 82.8 | 69.5 | 73.5                | 81.8 | 72.6 | 80.6 | 71.6 | 79.2 | 4.2                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 71.2                            | 124.3 | 76.6 | 70.2 | 120.1 | 75.9 | 69.3 | 116.4 | 75.1 | 38.6          | 82.1 | 37.7 | 80.8 | 36.8 | 79.3 | 78.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 10.6          | 8.8  | 7.8 | DB                         | 7.8 | 90.1               | 4.5 | 2.5                                                | 4.6 | 92.0       | 2.0 | 93.4       | -0.6 | 94.9       | -3.8 | 96.7 | (4) |
| (5)               |      |     | WB                         | 6.8 | 75.6               | 4.5 | 1.2                                                | 3.6 | 76.5       | 0.9 | 77.2       | -1.6 | 77.9       | -4.9 | 78.8 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 55.7        | 36.9  | 40.1  | 47.1  | 55.5  | 63.0  | 70.1  | 72.7  | 72.0  | 66.6  | 56.3  | 46.4  | 40.5  | (6)  |  |
|                                                                        | DBStd       | 14.44       | 9.81  | 8.65  | 8.49  | 7.04  | 6.09  | 3.78  | 3.34  | 3.30  | 5.17  | 7.49  | 7.81  | 8.41  | (7)  |  |
|                                                                        | HDD50       | 1398        | 418   | 292   | 159   | 28    | 2     | 0     | 0     | 0     | 0     | 27    | 158   | 314   | (8)  |  |
|                                                                        | HDD65       | 4173        | 872   | 697   | 554   | 289   | 106   | 7     | 1     | 2     | 43    | 284   | 559   | 759   | (9)  |  |
|                                                                        | CDD50       | 3471        | 11    | 15    | 70    | 193   | 406   | 603   | 703   | 682   | 497   | 221   | 50    | 20    | (10) |  |
|                                                                        | CDD65       | 773         | 0     | 0     | 0     | 5     | 46    | 160   | 238   | 219   | 91    | 13    | 1     | 0     | (11) |  |
|                                                                        | CDH74       | 5196        | 0     | 1     | 10    | 125   | 423   | 1127  | 1517  | 1293  | 604   | 92    | 4     | 0     | (12) |  |
|                                                                        | CDH80       | 1137        | 0     | 0     | 0     | 9     | 58    | 252   | 389   | 304   | 114   | 11    | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 3.2         | 3.9   | 4.1   | 4.1   | 3.9   | 3.1   | 2.7   | 2.3   | 2.2   | 2.3   | 2.7   | 3.3   | 3.5   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 59.0        | 4.9   | 4.6   | 5.4   | 4.7   | 4.3   | 5.6   | 5.5   | 5.6   | 5.2   | 3.8   | 4.2   | 5.3   | (15) |  |
|                                                                        | PrecMax     | 84.6        | 11.7  | 9.3   | 10.1  | 8.7   | 10.0  | 10.6  | 14.2  | 10.1  | 19.3  | 8.8   | 8.6   | 10.2  | (16) |  |
|                                                                        | PrecMin     | 39.7        | 1.3   | 1.9   | 2.0   | 2.1   | 1.5   | 1.6   | 1.9   | 1.0   | 0.9   | 0.3   | 1.5   | 2.3   | (17) |  |
|                                                                        | PrecStd     | 10.1        | 2.5   | 1.8   | 1.9   | 1.6   | 2.1   | 2.2   | 2.7   | 2.2   | 3.5   | 2.2   | 1.8   | 1.9   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 66.2  | 69.7  | 75.3  | 81.3  | 84.9  | 88.3  | 89.1  | 89.1  | 86.3  | 80.7  | 73.3  | 67.4  | (19) |  |
|                                                                        |             | MCWB        | 55.4  | 57.7  | 58.0  | 62.0  | 66.1  | 69.7  | 72.0  | 70.8  | 67.5  | 67.8  | 59.1  | 56.8  | (20) |  |
|                                                                        | 2%          | DB          | 61.5  | 64.2  | 71.1  | 77.7  | 81.5  | 85.4  | 86.4  | 86.1  | 83.3  | 76.5  | 68.4  | 63.4  | (21) |  |
|                                                                        |             | MCWB        | 54.2  | 54.1  | 56.2  | 61.3  | 65.5  | 69.4  | 71.6  | 71.2  | 67.9  | 63.3  | 56.0  | 56.1  | (22) |  |
|                                                                        | 5%          | DB          | 57.7  | 60.4  | 67.1  | 74.5  | 78.8  | 83.1  | 84.4  | 83.6  | 80.7  | 72.8  | 64.7  | 59.1  | (23) |  |
|                                                                        |             | MCWB        | 50.6  | 51.7  | 54.6  | 59.0  | 64.6  | 69.0  | 71.3  | 70.8  | 67.6  | 62.0  | 55.9  | 53.2  | (24) |  |
|                                                                        | 10%         | DB          | 53.1  | 55.8  | 62.8  | 71.0  | 76.0  | 80.4  | 82.1  | 81.0  | 77.8  | 69.3  | 61.6  | 55.2  | (25) |  |
|                                                                        |             | MCWB        | 48.1  | 47.4  | 52.4  | 57.6  | 63.8  | 68.1  | 70.8  | 70.1  | 66.9  | 60.7  | 53.8  | 50.1  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 59.2  | 60.4  | 61.0  | 67.8  | 69.8  | 73.4  | 75.0  | 75.1  | 72.1  | 70.2  | 64.7  | 62.2  | (27) |  |
|                                                                        |             | MCDB        | 63.6  | 66.3  | 70.3  | 73.7  | 79.2  | 82.2  | 84.2  | 83.1  | 79.4  | 77.2  | 67.3  | 65.1  | (28) |  |
|                                                                        | 2%          | WB          | 55.8  | 57.4  | 59.0  | 64.0  | 68.3  | 72.1  | 73.5  | 73.6  | 70.8  | 67.5  | 60.8  | 58.2  | (29) |  |
|                                                                        |             | MCDB        | 59.9  | 62.5  | 66.9  | 72.9  | 77.0  | 80.9  | 82.3  | 81.3  | 78.2  | 72.9  | 64.8  | 61.4  | (30) |  |
|                                                                        | 5%          | WB          | 52.4  | 53.2  | 57.1  | 61.8  | 67.0  | 70.9  | 72.7  | 72.6  | 69.5  | 65.2  | 58.7  | 54.7  | (31) |  |
|                                                                        |             | MCDB        | 55.7  | 57.7  | 63.7  | 69.8  | 75.0  | 79.5  | 81.2  | 79.8  | 76.1  | 69.6  | 62.9  | 57.6  | (32) |  |
|                                                                        | 10%         | WB          | 48.7  | 48.9  | 54.8  | 59.7  | 65.7  | 69.7  | 71.7  | 71.5  | 68.3  | 62.9  | 55.4  | 50.8  | (33) |  |
|                                                                        |             | MCDB        | 52.1  | 54.1  | 60.8  | 67.1  | 73.0  | 77.6  | 79.6  | 78.1  | 74.4  | 67.1  | 59.4  | 54.2  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 20.0  | 20.4  | 21.6  | 23.3  | 21.3  | 20.1  | 18.4  | 18.2  | 19.1  | 21.6  | 23.2  | 18.9  | (35) |  |
|                                                                        |             | MCDBR       | 27.1  | 28.4  | 28.6  | 30.3  | 26.3  | 23.9  | 21.7  | 22.2  | 24.0  | 25.6  | 28.2  | 23.6  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 18.1  | 18.2  | 15.9  | 14.9  | 12.2  | 10.1  | 9.1   | 9.2   | 10.5  | 13.5  | 17.0  | 16.3  | (37) |  |
|                                                                        |             | MCWBR       | 20.5  | 22.6  | 22.7  | 23.4  | 20.8  | 20.1  | 18.9  | 18.8  | 18.8  | 18.0  | 19.8  | 19.0  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.295 | 0.303 | 0.334 | 0.373 | 0.427 | 0.453 | 0.493 | 0.491 | 0.387 | 0.353 | 0.314 | 0.292 | (40) |  |
|                                                                        |             | taud        | 2.543 | 2.511 | 2.454 | 2.343 | 2.209 | 2.170 | 2.085 | 2.072 | 2.349 | 2.439 | 2.514 | 2.579 | (41) |  |
|                                                                        | Ebn at Noon |             | 291   | 300   | 298   | 289   | 273   | 265   | 253   | 251   | 276   | 277   | 280   | 285   | (42) |  |
|                                                                        |             | Edh at Noon | 25    | 29    | 33    | 39    | 46    | 47    | 51    | 51    | 37    | 31    | 26    | 23    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 822   | 1021  | 1351  | 1674  | 1840  | 1936  | 1802  | 1687  | 1462  | 1219  | 918   | 708   | (44) |  |
|                                                                        | RadStd      |             | 51    | 110   | 86    | 127   | 137   | 135   | 125   | 79    | 131   | 126   | 80    | 64    | (45) |  |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            |  | +0.95 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +256  | +148  |
| Regional (41 neighbors) |  | +0.97 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +277  | +180  |

Nomenclature: See separate page